

Sanjay J K

Computer Science Student @ VIT — AI Systems & Full Stack Developer
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SUMMARY

Focused on building scalable AI systems and real-world intelligent applications. Computer Science student at VIT (CGPA 9.34) building **AI systems and scalable full-stack applications**. Experience developing **LLM pipelines, RAG systems, and multi-agent architectures**; filed **two AI patents** and selected for UKIERI-SPARC international research program.

SKILLS

Languages: Python, C++, Java, C, JavaScript

Frameworks & Web: React, Next.js, Node.js, Express.js, REST APIs, HTML, CSS, TailwindCSS

Databases: MongoDB, PostgreSQL, SQL, ChromaDB

AI / Machine Learning: PyTorch, Scikit-learn, Retrieval-Augmented Generation (RAG), Prompt Engineering

Tools: Git, GitHub, LangChain, LangGraph, Langfuse, Ollama

EXPERIENCE

Frontend Developer Intern — Valsco Technology (Remote)

Jun 2024 – Sep 2024

- Developed **NoteSwap**, a React-based platform used by **200+ students** to exchange academic resources.
- Improved frontend performance through component optimization and lazy loading, reducing page load time by **20%**.
- Designed responsive UI components and improved usability across desktop and mobile interfaces.

PROJECTS

LLM Knowledge Assistant (Production RAG System)

Python, LangChain, ChromaDB

- Built a **Retrieval-Augmented Generation system** supporting ingestion of PDFs, Markdown and web content.
- Implemented **hybrid retrieval** using vector search + BM25 with reciprocal rank fusion.
- Added cross-encoder reranking and evaluation pipelines with **Langfuse observability**.

InterviewAI — Real-Time Voice AI Interviewer

FastAPI, React, WebSockets, Ollama

- Built a **real-time voice AI interviewer** with Deepgram ASR, Ollama LLMs, and ElevenLabs TTS, achieving **sub-1.2s end-to-end latency** through an event-driven streaming architecture.
- Developed an **interruption-aware conversational engine** with asynchronous speech debouncing, stream cancellation, and full-duplex WebSocket orchestration for natural interview interactions.
- Implemented **session telemetry and analytics dashboards** tracking ASR, LLM, and TTS latency metrics, with automated feedback generation and exportable interview reports.

Neo4j GraphReasoner — NLP Knowledge Graph

Neo4j, Node.js, Ollama

- Built **interactive research graph** with multi-hop Cypher reasoning and physics-based visualization.
- Developed **LLM-powered query system** translating natural language to Cypher with reasoning traces.
- Integrated **end-to-end observability** (Langfuse, tracing, alerts) for queries and LLM pipelines.

SwarmIQ — Social Simulation Engine (GraphRAG + LLMs)

Python, Ollama, NetworkX

- Built **agent-based simulation system** modeling opinion dynamics using local LLMs and GraphRAG.
- Designed **cognitive simulation loop** with memory, influence propagation, and emergent behavior modeling.
- Implemented **knowledge graph pipeline** from documents → entities → relationships → simulation world.

PATENTS

- Published (2026, Indian Patent Office):** Pan-Chronological Vision–Language Transformer for Cross-Era Tamil Script Decipherment.
- Published (2026, Indian Patent Office):** Personalized AI System for Color Vision Deficiency using Perceptual Gamut Modeling and Adaptive Recommendation.

EDUCATION

Vellore Institute of Technology (VIT) — B.Tech Computer Science

2023 – 2027

CGPA: **9.34**

CBSE High School

Score: **93.5%**

ACHIEVEMENTS & LEADERSHIP

- GATE CSE 2026:** AIR **1603** (Score: **670**).
- UKIERI Research Scholar (2025) – Heriot-Watt University, UK:** Selected among **9 students nationally**.
- Judges' Choice Award – Sankalp Innovation Challenge 2026 (MNNIT Allahabad):** Built **JanVedha AI**, a civic decision intelligence platform.
- Finalist – Google Cloud Agentic AI Hackathon 2025:** Built *ShikshaMitrah*, an AI assistant for rural educators.
- Finalist – India Innovates 2026 (Bharat Mandapam, New Delhi):** Selected among top teams at India's largest civic-tech hackathon.
- Finalist – DeepShiva Open Source LLM Hackathon (Healthcare):** Ranked among the **top 3 teams nationally**.
- Board Member – Design Lead, AEE VIT Chapter & Senior Core Member – CodeChef VIT & TAM-VIT**